

Mediterranean alpha-linolenic acid-rich diet in secondary prevention of coronary heart disease.

In a prospective, randomised single-blinded secondary prevention trial we compared the effect of a Mediterranean alpha-linolenic acid-rich diet to the usual post-infarct prudent diet. After a first myocardial infarction, patients were randomly assigned to the experimental (n = 302) or control group (n = 303). Patients were seen again 8 weeks after randomisation, and each year for 5 years. The experimental group consumed significantly less lipids, saturated fat, cholesterol, and linoleic acid but more oleic and alpha-linolenic acids confirmed by measurements in plasma. Serum lipids, blood pressure, and body mass index remained similar in the 2 groups. In the experimental group, plasma levels of albumin, vitamin E, and vitamin C were increased, and granulocyte count decreased. After a mean follow up of 27 months, there were 16 cardiac deaths in the control and 3 in the experimental group; 17 non-fatal myocardial infarction in the control and 5 in the experimental groups: a risk ratio for these two main endpoints combined of 0.27 (95% CI 0.12-0.59, $p = 0.001$) after adjustment for prognostic variables. Overall mortality was 20 in the control, 8 in the experimental group, an adjusted risk ratio of 0.30 (95% CI 0.11-0.82, $p = 0.02$). An alpha-linolenic acid-rich Mediterranean diet seems to be more efficient than presently used diets in the secondary prevention of coronary events and death.